Detecting Cognitive States from Degraded EEG Signals:

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Abstract

Cognitive state detection using electroencephalography (EEG) has shown significant promise in laboratory settings, yet the vast majority of published studies evaluate models only on clean, preprocessed recordings. This leaves an important practical question unanswered: can trained models maintain predictive validity under the signal degradation conditions common in real-world deployments, such as electrode loss, environmental noise, truncated sessions, and corrupted preprocessing? This project addresses that gap using the DEAP affective dataset[1], which comprises EEG recordings from 32 participants across 40 emotionally evocative music video trials, captured at 128 Hz across 32 channels. Six classifiers SVM, Random Forest, LDA, CNN, LSTM, and a CNN-LSTM hybrid were trained and fused into a weighted ensemble. Arousal and valence labels were binarized at dataset medians of 3.315 and 3.08, respectively, to map cognitive states. Under clean conditions, the ensemble achieved 66.7% test accuracy. When four degradation conditions were introduced systematically Gaussian noise injection, channel dropout (8 of 32 electrodes zeroed), truncated recording windows (last 40% zeroed), and corrupted feature matrices (30% of channel-band values distorted) accuracy remained between 63–65% in three of four scenarios. The most severe drop occurred under electrode failure, yielding 56.8%, a 9.8 percentage point decline that nonetheless remained above the 50% chance baseline. Predictions were then mapped onto Russell's Circumplex Model of Affect to produce four interpretable cognitive states: Positive, Stressed, Fatigued, and Calm. These results provide an empirically grounded robustness profile for EEG-based cognitive state recognition and offer concrete design guidance for practitioners developing brain-computer interface systems.
Keywords: EEG, DEAP dataset, arousal classification, cognitive states, signal degradation, weighted ensemble, CNN, LSTM, Russell's Circumplex Model, brain-computer interfaces

I. Introduction

EElectroencephalography is the most common modality of non-invasive brain-computer interfaces since it captures the overall electrical activity of groups of cortical neurons at milliseconds time resolution. EEG hardware is cheap and can be used in a portable way as compared to functional MRI or PET imaging, which are not always ambulatory. These features have rendered EEG to the focus of emotion recognition, cognitive workload, driver fatigue, and clinical neurological evaluation studies [9]. These features have rendered EEG the center of emotion recognition, cognitive workload, driver fatigue, and clinical neurological assessment research studies. In spite of this scope of usage, the field is marked by a certain methodological tension. Published accuracy measures are regularly derived based on carefully preprocessed, artifact-filtered measurements which are measured under idealized laboratory conditions. The difference between these figures and performance which can be attained in real life is hardly ever measured. Practically, a 32-channel EEG headset has to compete with head motion artifact which corrupts the signal with broadband noise, electrodes which lose gel contact and become flat or saturated channels and software pipelines which can inject silent corruptions during feature extraction. When such degraded data is fed into a model that was trained on clean data, there is no guarantee that the model will be robust.

That gap inspired this project. The study question that guides the study is: *can EEG-based cognitive state detection remain reliable when input signals are noisy, spatially incomplete, temporally truncated, or feature-corrupted?* To answer it rigorously, six classifiers based on both classical and deep learning paradigms were trained on DEAP and evaluated on four regimes of degradation independently constructed on the held-out test partition. Then ensemble predictions were run with ground-truth valence labels and projected using a Circumplex Model developed by Russell to produce four categories of cognitive state of practical interpretive value. The rest of this report will be structured in the following way. Section II formalizes the problem. Section III discusses the related literature. The dataset, preprocessing pipeline, feature extraction strategy, model architectures, and ensemble construction, as well as the robustness testing protocol, are described in Section IV. All quantitative results are provided and analyzed in Section V. Section **??** describes the contribution of the project to the existing work. Section VI concludes and provides directions to future research.

II. Problem Statement

Binary classification of arousal on DEAP is not that easy. Subject-independent models even in clean conditions typically are in the 65-75% percent accuracy range [1], due to EEG morphology differences which are large between individuals as a function of skull thickness, electrode impedance variability and individual neuroanatomical variations in cortical folding behavior. The issue that this project will deal with has prolonged that base challenge in two right angles.

The initial one is the *model diversity*. Previous research on DEAP has largely tested a single or two architectures individually, complicating the ability to ascribe performance to architecture decisions as opposed to hyperparameter tuning or choice of data split. A six model test on the same splits is a more controlled basis of architectural inference. The second and the main one is the *robustness under degradation*. There are four signal corruption conditions under investigation:

1) **Gaussian noise (30% of signal standard deviation)**: simulates the noise of the environment (interference of electromagnetic phenomena and muscle movements) and artifacts of muscle movements that pollute all channels equally.
2) **Missing channels (8 of 32 electrodes zeroed)**: Simulates a hardware failure, bad electrode-scalp contact or an impedance-induced dropout and causes 25% of the spatial information to be lost permanently.
3) **Incomplete recording (last 40% of window zeroed)**: It imitates recording interruption, overflow of buffers or truncation of sessions at the 60 percent mark of a 4-second analysis window.
4) **Distorted input (30% of channel-band matrix corrupted)**: The input to the CNN is distorted by simulating errors at the feature extraction phase, such as numerical instabilities, pipeline bugs or memory corruption.

No previous DEAP experiment has assessed a multi-model ensemble in all four conditions of degradation on the identical sealed test set. This is the main input and the methodological essence of this project..

III. Related Work

a The DEAP Dataset

Queen Mary University of London collected DEAP and it is still the most popular benchmark of EEG-based affective computing [1], [2]. Thirty-two healthy subjects viewed 40 1-minute music videos with 32-channel EEG being recorded at 128 Ḣz. Following each video clip, the participants rated themselves on a continuous basis on four dimensions: valence, arousal, dominance, and liking. The resulting data gives a naturalistic stimulus paradigm in terms of which the music videos have

an ecologically valid inducers of emotion, a rich multi-channel EEG recording and equal subjective ratings.

b EEG Feature Representations

The leading feature engineering method to EEG classification is band power estimation through the periodogram of Welch [5]. A 160-dimensional feature vector by averaging estimates of power spectral density across all 32 channels across the canonical frequency bands, delta (1-4 Ḣz), theta (4-8Ḣz), alpha (8-13Ḣz), beta (13-30Ḣz) and gamma (30-45Ḣz), is obtained, which captures the oscillatory dynamics most correlated with change of cognitive state. The beta power specifically has been consistently associated with alert and active mind states, and the arousal has been correlated with alpha power inhibition [9].

Later, spatial representations, which maintain the topographic layout of electrodes, have been experimented with. Two-dimensional input matrices enabling convolutional filters to learn which channel and frequency band pairs are co-activated have been demonstrated to be better than flat feature vectors in arousal and valence classification tasks [6].

c Deep Learning for EEG Classification

Lawhern et al. proposed EEGNet, a small convolutional network that is particularly EEG-specific, which separates temporal convolution and spatial filtering of EEGs into two functions [13]. Later studies have used both LSTM and CNN structures in conjunction with DEAP and shown that both frequency-domain features and temporal modeling of raw EEG are complementary. Li et al. suggest a hierarchical spatial-temporal network, which models the relationships between regions and globes explicitly through the model of electrodes [6]. Regardless of these innovations, there is little literature on robustness testing in the presence of controlled signal degradation..

d Ensemble Methods

Machine learning has a long-standing tradition of ensemble aggregation of heterogeneous classifiers [12]. One principled approach is to weighted average the probability outputs (with weights proportional to validation accuracy) which enables more reliable models to have greater influence on the final prediction with no extra training. In situations where structurally distinct inductive biases are encoded by the constituent models, such as when a kernel SVM is used on frequency-domain features and a CNN is used on a spatial matrix, their errors are likely to be uncorrelated, and the ensemble is more likely to cover the instance space complementarily.

e Russell's Circumplex Model

The Circumplex Model of Affect by Russell is a two-dimensional model of emotional states based on arousal (between calm and excited) and valence (between unpleasant and pleasant) on a plane of affect [3]. The framework offers a psychologically based, principled basis of mapping binary arousal and valence predictions on to named cognitive states. It has been used broadly in affective computing to encode classifier results in meaning derived that are relevant to end users and domain practitioners.

IV. Methodology

a Dataset Description

The DEAP dataset [1] is a collection of EEG recordings of 32 participants (s01.dat to s32.dat). Each of the subjects performed 40 trials, with each being associated with a separate one-minute music video stimulus. EEG was sampled at 128 Hz and 32 electrodes on the scalp. Following each test, self-report ratings were given by the participants on arousal and valence scales. In this project, the number of subjects (32) and number of trials per subject (40) were employed which gave maximum coverage to training. The data median of 3.315 was used to binarize arousal labels (above the median is considered High Arousal, below it is Low Arousal). The valence scores were kept continuous and then thresholded to 3.08 to map cognitive states. Table I gives a summary of the most important parameters of the dataset configuration.

TABLE I: Dataset and Pipeline Configuration Parameters

Parameter	**Value**
Subjects	32
Trials per subject	40
EEG channels	32
Sampling rate	128 Hz
Arousal binarization threshold	3.315 (dataset median)
Valence threshold	3.08
Window length	4 s (512 samples)
Window overlap	50% (STEP = 256 samples)
Total windows (pre-balancing)	38,400
Windows per class (post-balancing)	19,200

The visualization includes four panels: (top-left) distribution of classes after balancing, (top-right) raw EEG signals of the samples in representative channels, (bottom-left) comparison of mean band power by arousal class, and (bottom-right) band-power correlation matrix. These findings validate data balancing, signal quality and existence of discriminative spectral organization before model training. The exploratory data analysis of the DEAP data. *Top left*: class distribution after random undersampling - 19,200 windows per class, perfectly balanced. *Top right*: sample raw EEG across five channels over a 4-second window; the large-amplitude spike on channel 4 illustrates the artifact morphology that the bandpass filter is designed to suppress. *Bottom left*: mean band power per arousal class across all five frequency bands; High Arousal consistently exhibits greater power, with the largest separability in the beta band. *Bottom right*: band power correlation matrix showing moderate correlation between adjacent bands (delta-theta, alpha-beta) while confirming that each band retains independent discriminative information.

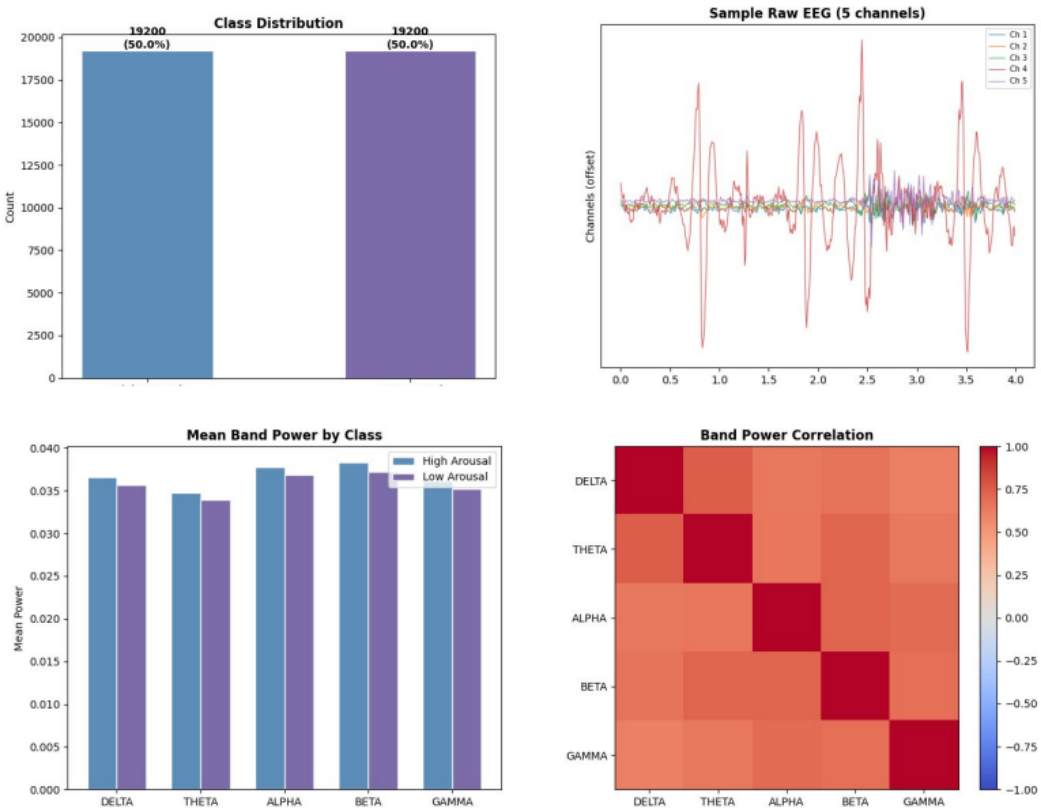

Fig. 1: Exploratory Data Analysis of the DEAP Dataset

b Signal Preprocessing

EEG data were first converted to raw and then subjected to 2-stage processing before any feature extraction or model training.

b1 Bandpass Filtering

A Butterworth bandpass filter of order 5 with cutoff frequencies of 1Hz and 45 Hz was used with the `scipy.signal.filt` a zero-phase filter, which does not add any time-domain distortion. The 1 Hz low cutoff removes DC offset and slow electrode drift artifacts. The upper cutoff frequency of 45 Hz reduces high-frequency electromyographic contamination due to the activity of facial and scalp muscles. To maintain phase congruency across channels, which is pertinent to the spatial coactivities that the CNN will need to learn, zero-phase filtering was selected over causal filtering.

b2 Per-Channel Z-Score Normalization

The channels were individually normalized, by replacing the mean in time, and dividing it by the standard deviation (with a small epsilon of 10^{-8} added for numerical stability):

$$\hat{x}_{c,t} = \frac{x_{c,t} - \mu_c}{\sigma_c + 10^{-8}}$$

Normalization per-channel is important since amplitude of EEG differs significantly among subjects due to physiological factors (skull thickness, cortical geometry, electrode impedance). Global normalization would confound amplitude change with cognitive state change and, as a result, models would unintentionally learn subject-specific differences than actual arousal signatures.

c Windowing and Segmentation

The 60-second trial was subdivided into 4-second (512 samples at 128 Hz) analysis windows every 2 seconds (256 samples), giving an overlap of 50%. The duration of the 4-second window was chosen to ensure several full cycles were resolved even at the lowest frequency band of interest (delta at 1 Hz) but was kept short to ensure it was compatible with near-real-time applications. The 50%percent overlap actually doubles the training sample size without having to incur extra data collection expenses. High and Low Arousal windows were observed to be a little bit unbalanced; random undersampling was used to get both classes to precisely 19200 windows each, so that none of the accuracy measures are artificially inflated by the imbalance between the two classes.

d Feature Extraction

The preprocessed windows were then assembled into three different representations of the features, each a feature representation specific to the inductive bias of a given family of models.

d1 Band Power Features (160-dimensional)

In the case of SVM, Random Forest, and LDA, band power was calculated with Welch periodogram technique of band power computation as given by Welch in 1967 with a segment of 256 samples long [5]. The mean power spectral density of each 32 channels was calculated in each of the five canonical frequency bands resulting in a $5 \times 32 = 160$-dimensional feature vector per window yielding. This model is small, easily interpretable and directly related to known neurophysiological correlates of arousal.

d2 Channel-Band Matrix (32×10)

In the case of the CNN, a two-dimensional matrix was designed where rows are the 32 EEG channels and columns are the ten features (the five band powers and five clinically motivated spectral ratios (theta/alpha drowsiness, beta/alpha cognitive activation, theta/beta attention, (theta+alpha)/beta engagement index, and gamma/alpha)). This representation maintains the spatial arrangement of the electrode montage, which allows convolutional filters to learn spatially localized patterns of co-activation between channels and frequency bands.

d3 Raw EEG Time Series (32×512)

In the case of LSTM and CNN-LSTM hybrid, the time series in its preprocessed form was used and spectral features were not extracted. The shape matrix of size 32×512 was permuted to 512×32in advance of the recurrent layers such that the time dimension acts as the sequence dimension of the LSTM. All the temporal dynamics are retained by this representation that frequency-domain averaging eliminates.

e Data Partitioning

Stratified sampling was used to split the dataset into three non-overlapping subsets and ensure that the class proportions are the same in all splits. A quarter of the data ($\approx 7{,}680$ windows per class) was held as a closed test set and was not used until end assessment. The 80% was further split into training set (practically 68% and a validation set (($\approx 15\%$ of total). The entire normalization and scaling values (StandardScaler metrics of the classical models, channel-band matrix means and standard deviations of the CNN) were only calculated on the training set and applied to the validation and test partitions in the same way, eliminating any data leakage.

f Model Architectures

Figure 2 shows the architecture of the entire system and how it interacts with preprocessing, feature extraction, model diversity, and ensemble aggregation. The general architecture of the entire EEG processing pipeline. The pipeline starts with raw EEG acquisition of 32 channels, and then preprocesses (bandpass filter, and normalization), cuts the signal into overlapping windows and derives three feature representations: band power, channel-band matrix, and raw EEG sequence. They are run through six models (SVM, Random Forest, LDA, CNN, LSTM, and CNN-LSTM hybrid), the results of which are aggregated through a weighted ensemble approach. The Circumplex Model is a map of predictions of final states to cognitions states [3]. Figure 2 shows the architecture of the entire system and how it interacts with preprocessing, feature extraction, model diversity, and ensemble aggregation.

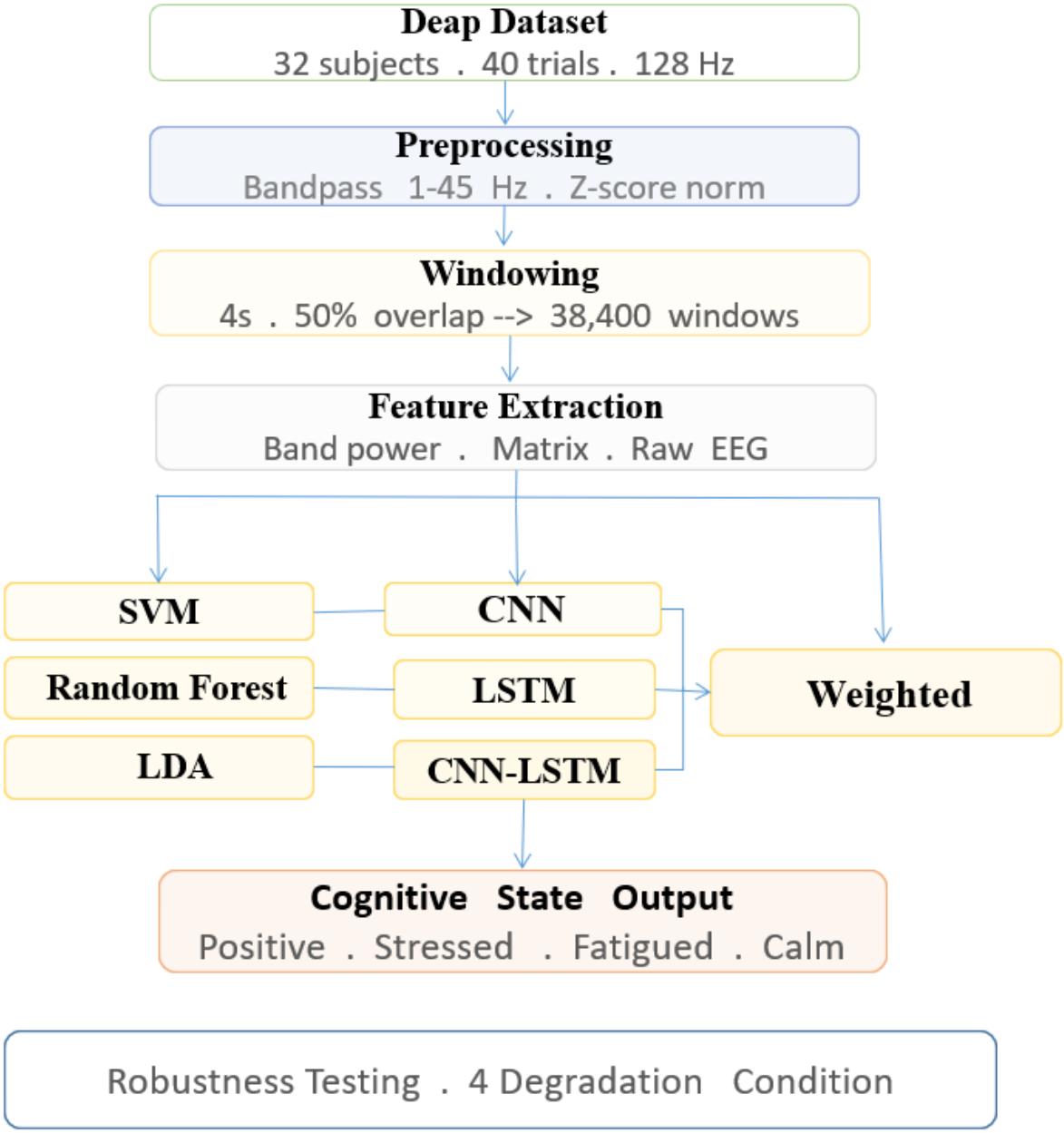

Fig. 2: Proposed Multi-Model Ensemble Architecture

g System Architecture Overview

The architecture presented in Figure 2 is created as a multi-stage (robust) pipeline of EEG-based cognitive state detection in both ideal and degraded signal conditions. The process commences with the dataset of DEAP, which contains multi-channel recordings of EEG under controlled conditions of experiment. Such signals are initially preprocessed, such as bandpass filtering (1-45 Hz) to remove physiological and environmental noise, and Z-score normalization to bring signal amplitude across channels and subjects to a common level. The continuous EEGs are then divided into overlapping 4-second long time windows with 50% overlap after preprocessing. The windowing approach scales up the effective dataset size but maintains the temporal dynamics that are needed in downstream modelling.

The extraction of the features is then carried out in order to obtain three complementary representations. Band power models capture frequency-domain attributes and are inputted by classical machine learning models like SVM, Random Forest and LDA. The CNN uses a structured channel-band matrix to learn spatial patterns of activation in response to the electrode to capture spatial relationships. Simultaneously, raw EEG time-series data is stored to sequence-based models (LSTM and CNN-LSTM), which allows capturing of time-dependences. The six models are independent and each one of them exploits a unique inductive bias. Classical models emphasize on statistical separability whereas deep learning models learn fine spatial and temporal dynamics. Their results are combined with a weighted ensemble method, whereby model results are weighted based on their validation results. This enhances overall predictive accuracy, and addresses shortcomings of individual models.

The end ensemble result is then mapped to cognitive states based on the Circumplex Model proposed by Russell, which give interpretative categories, namely, Positive, Stressed, Fatigued, and Calm. In order to assess the real-world applicability, robustness testing is conducted in four degradation scenarios, such as noise injection, missing channels, incomplete recordings, and corruption of features. This architecture is not only highly predictive, but also resistant to real-life issues that are faced when deploying EEG in practice.

g1 Support Vector Machine

A SVM fitted with RBF kernel (C=10, gamma=scale) was fitted on the standardized 160-dimensional band power features. The training was done on a stratified subsample of the training set of 15% due to the computational tractability; training on 26,000 or more samples is forbidden. RBF kernel was selected due to the fact that the boundary of arousal decision in 160-dimensional band-power space is non-linear. Platt scaling was used to obtain probability outputs.

g2 Random Forest

sAn ensemble of 200 decision trees (Random Forest) was trained on the entire 160-dimensional set of features without further standardization as the tree-based methods are also scale-invariant. The analysis of feature importance allowed asserting that beta- and gamma-band power at frontal and temporal electrode sites were always the most discriminative features.

g3 Linear Discriminant Analysis

LDA identifies the single linear projection which maximizes the ratio of between class scatter to within class scatter. It was added as a fast, interpretable baseline, having a theoretically inspired relation to Gaussian class-conditional distributions. Its performance was supposed to be the worst among the three classical models because the arousal decision boundary is not a globally linear decision boundary in the band-power space.

g4 Convolutional Neural Network

The CNN uses the channel-band matrix of size: $32 \times 10 \times 1$. It has three Conv2D blocks, each having depths of 64, 128, and 256; they are complemented by Batch Normalization and a

Dropout layer (0.25 in the first two, 0.35 in the third). To minimize the number of parameters in an effort to enhance generalization, Global Average Pooling is used in place of flattening. The Softmax output is preceded by two fully connected layers (256 and 128 units) with intermediate Dropout. The model consists of 470,402 parameters to be trained. The Adam optimizer (learning rate 10^{-3}) and categorical cross-entropy loss were used, as well as early stopping (patience 12) and ReduceLROnPlateau (factor 0.5.

g5 Long Short-Term Memory Network

The LSTM uses the raw EEG matrix of dimensions 32×512 , and is permuted to 32×512 in such a way that the sequence dimension is the temporal position. It is followed by two stacked LSTM layers (128 units that give back sequences, and then 64 units that give back sequences) and a Dense layer (128 units, ReLU) with Dropout (0.3) and Softmax output. There are 140,418 trainable parameters in the model. The LSTM learns to capture EEG pattern dynamics in the 4-second window; data that spectral averaging removes; whereas the band-power classifiers are structurally complementary to the LSTM.

g6 CNN-LSTM Hybrid

The hybrid model operates two parallel streams. The CNN stream uses the channel-band matrix with 32×10 channels and passes it through two Conv2D blocks (64 and 128 filters) and then Global Average Pooling, and a final Dense(128) layer. The raw EEG of size 3 32×512 is handled by the LSTM stream using the same architecture of two-layers LSTM as the solitary LSTM. Both streams generate 128 dimensional representations which are concatenated and fed to Dense layers (256 and 128 units) and finally to Softmax classification. This model has 330, 882 trainable parameters. This architecture expressly captures spatial frequency-band coactivation and temporal signal dynamics in a single prediction. A table of all model configurations appears in Table II.

TABLE II: Model Architecture Summary

Model	**Input**	**Parameters**	**Key Inductive Bias**
SVM (RBF, C=10)	Band power (160-dim)	—	Non-linear margin maximization
Random Forest (200 trees)	Band power (160-dim)	—	Ensemble of decision boundaries
LDA	Band power (160-dim)	—	Linear class-conditional projection
CNN	Channel-band (32×10)	470,402	Spatial co-activation patterns
LSTM	Raw EEG (32×512)	140,418	Temporal signal dynamics
CNN-LSTM Hybrid	Dual-stream	330,882	Spatial + temporal fusion

h Weighted Ensemble Construction

Each of the six models produces a two-class probability vector for every test window. The ensemble aggregates these vectors via a weighted average:

$$\hat{p}(y \mid \mathbf{x}) = \sum_{k=1}^{6} w_k \cdot p_k(y \mid \mathbf{x}), \quad w_k = \frac{\mathrm{acc}_k^{\mathrm{val}}}{\sum_{j=1}^{6} \mathrm{acc}_j^{\mathrm{val}}}$$

where $\mathrm{acc}_k^{\mathrm{val}}$ denotes the validation set accuracy of model k. The final class prediction is the argmax of $\hat{p}$. This weighting scheme ensures that models with higher empirical reliability on the validation partition contribute proportionally more to the ensemble decision, without requiring any additional training.

i Cognitive State Mapping

Russell's Circumplex Model [3] organizes emotional experiences on a two-dimensional arousal-valence plane. Applying this framework to the ensemble outputs produces four mutually exclusive cognitive state labels:

- **Positive**: High Arousal prediction AND valence > 3.08 (engaged, excited)
- **Stressed**: High Arousal prediction AND valence ≤ 3.08 (alert but negatively valenced)
- **Fatigued**: Low Arousal prediction AND valence ≤ 3.08 (low energy, disengaged)
- **Calm**: Low Arousal prediction AND valence > 3.08 (relaxed, content)

Note that valence in this mapping is taken from DEAP ground-truth self-report labels, not predicted from EEG. A fully autonomous system would require a separate valence prediction head; this limitation is discussed in Section V.

j Robustness Testing Protocol

Each of the four degradation conditions was applied independently to the sealed test set after all model training was complete. This design ensures that no degradation-specific information could influence model weights.

V. Results and Discussion

a Individual Model Performance

Table III presents the accuracy on the test set of all six individual models and the ensemble. Random Forest was the most accurate with a score of 62.5%, then SVM with 60.9% and lastly LDA with 59.8%. It is not surprising that LDA performs worse than the other classical methods: the arousal decision boundary in the 160-dimensional band-power space is non-linearly structured, and cannot be described by a linear discriminant.

TABLE III: Test Accuracy on DEAP Held-Out Set

Model	**Accuracy**	**Parameters**	**Input Representation**
SVM (RBF, C=10)	60.9%	—	Band power (160-dim)
Random Forest	62.5%	—	Band power (160-dim)
LDA	59.8%	—	Band power (160-dim)
CNN	66.0%	470,402	Channel-band (32×10)
LSTM	63.1%	140,418	Raw EEG (32×512)
CNN-LSTM Hybrid	63.6%	330,882	Dual-stream
Weighted Ensemble	**66.7%**	**—**	**All three**

TThe CNN scored 66.0%, which is higher than that of LSTM (63.1%) and CNN-LSTM hybrid (63.6%). The CNN merit is explained by the fact that the spatial structure of arousal-related EEG: the power increase in the beta-band in frontal and temporal clusters of electrodes is the most reliable discriminative signal in DEAP, and the 2D convolutional filters on the channel-band matrix are formally structured to respond to exactly such spatially contiguous patterns. The LSTM acts to extract a real but more subtle signal: the arousal-related dynamics dynamics through the 4-second window based on the raw temporal sequence. That signal is not as reliably discriminative with a binary label based on a 1-minute self-report rating.

The fact that the hybrid cannot perform better than the CNN alone is instructive. The hybrid combines streams of two inductive biases in the same model, adding variance to the LSTM stream

that is not always beneficial, competing with the more dependable spatial predictions of the CNN. This result indicates that the advantage of architectural diversity may be more achieved at the aggregation level of the ensemble than the single multi-stream model.

The optimal overall result is the weighted ensemble at 66.7%. Its marginal gain relative to the CNN alone is due only to situations in which the CNN does not provide enough spatial features, but the classical models based on simpler band-power summaries occur by chance to be accurate- a literal application of the error-complementarity principle which drives the construction of ensemble.

Fig 3 The illustration shows at the*Top left*: test accuracy comparison across all six models and the ensemble, with 70% and 80% reference lines. *Top right*: ensemble confusion matrix showing 2,625 correct High Arousal predictions and 2,494 correct Low Arousal predictions, indicating balanced classification performance. *Bottom left*: CNN vs. LSTM training curves across epochs, where the LSTM converges quickly but plateaus around 62%, while the CNN continues improving to higher accuracy. *Bottom right*: cognitive state distribution on the test set, with all four states (Positive, Stressed, Fatigued, Calm) meaningfully represented. Figure a presents the overall performance of all models and the ensemble, including classification accuracy, training dynamics, and cognitive state outcomes

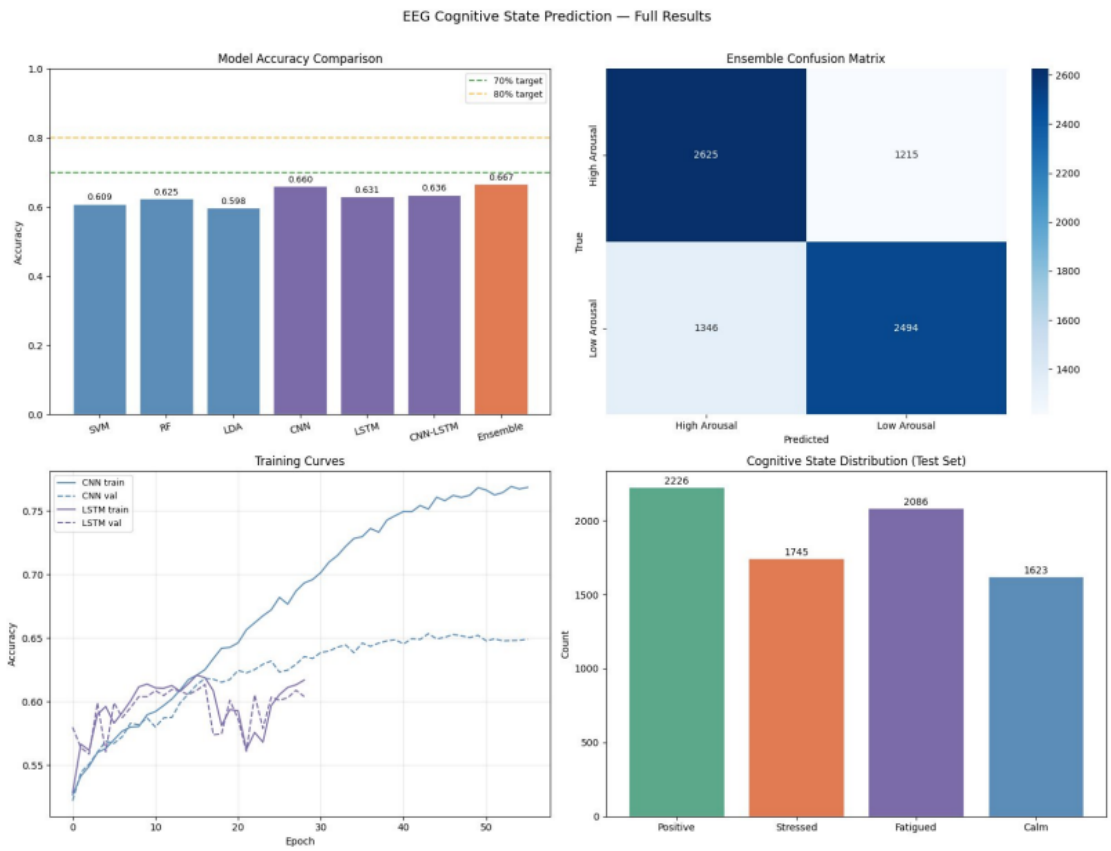

Figure 3: Complete Model result

b Confusion Matrix Analysis

The ensemble confusion matrix (Figure a, top right) shows 2,625 correct High Arousal predictions and 2,494 correct Low Arousal predictions, with 1,215 High Arousal windows misclassified as Low and 1,346 Low Arousal windows misclassified as High. The model identifies High Arousal more reliably, consistent with the EDA finding that High Arousal windows exhibit visibly elevated

beta-band power - a clean, consistent separating feature. Low Arousal windows show greater spectral overlap with High Arousal in the lower frequency bands, and this overlap is the primary source of misclassification. Critically, the confusion is approximately symmetric: precision and recall are comparable for both classes, indicating that the ensemble is not exploiting a systematic bias toward one label.

c Training Dynamics

The training curves (Figure a, bottom left) reveal markedly different learning trajectories for the CNN and LSTM. The LSTM reaches near-plateau validation accuracy of 61–62% within 10–15 epochs, suggesting that the temporal dynamics it captures are quickly encoded but have limited ceiling. The CNN's validation accuracy continues rising through epoch 20–30, reaching 65% by epoch 50, with the training curve rising above that. The gap between CNN training and validation accuracy reflects the cross-subject generalization difficulty inherent in DEAP rather than classical overfitting: the model is learning genuine EEG patterns, but those patterns are partially subject-specific. Dropout, early stopping (patience 12), and class weighting prevented both models from diverging.

d Robustness Under Signal Degradation

Figure 3 and Table IV present the central empirical finding of this project.

TABLE IV: Ensemble Accuracy Under Degradation Conditions (DEAP Test Set)

Condition	**What Degrades**	**Accuracy**	**Drop**
Clean baseline	Nothing	66.7%	—
Gaussian noise (30%)	All features (signal-level)	63.4%	3.2%
Missing channels (25%)	8 of 32 electrodes lost	56.8%	9.8%
Incomplete rec. (60%)	Last 40% of window zeroed	65.0%	1.7%
Distorted input (30%)	Channel-band matrix only	63.1%	3.5%

Gaussian noise at 30% of signal standard deviation caused only a 3.2 percentage-point drop (66.7% $\rightarrow$ 63.4%). This resilience is attributable to the noise-averaging property of Welch's periodogram: random Gaussian noise distributes uniformly across all frequency bins, and the per-band averaging step attenuates its contribution to the 160-dimensional feature vector. The SVM, RF, and LDA-all operating on Welch-averaged features-are inherently resistant to this degradation mode.

Incomplete recording was the most benign condition, yielding only a 1.7 percentage-point drop (66.7% $\rightarrow$ 65.0%). Zeroing the last 40% of a 4-second window has minimal impact because arousal-relevant beta-band patterns stabilize within the first 2 seconds; the trailing 1.6 seconds carry redundant information for this binary classification task. This finding has a direct practical implication: brief recording interruptions are largely harmless, and systems need not discard windows with minor temporal gaps.

Distorted input affecting 30% of the channel-band matrix values caused a 3.5 percentage-point drop (66.7% $\rightarrow$ 63.1%). Because this corruption targets only the CNN's input representation, the SVM, RF, LDA, and LSTM predictions are unaffected. The ensemble weighting absorbs the CNN's reduced accuracy by relying proportionally more on the uncorrupted models—precisely the resilience that ensemble diversity is designed to provide.

Missing channels was the worst-case condition, causing a 9.8 percentage-point drop (66.7% → 56.8%). Eight zeroed channels permanently remove 25% of the spatial brain signal. Unlike Gaussian noise, which sits additively on top of a recoverable real signal, a zeroed channel contains no recoverable information: the feature-extraction step has nothing to work with. Despite this, the ensemble remains above the 50% chance baseline at 56.8%, indicating that it degrades gracefully rather than failing catastrophically under worst-case hardware failure.

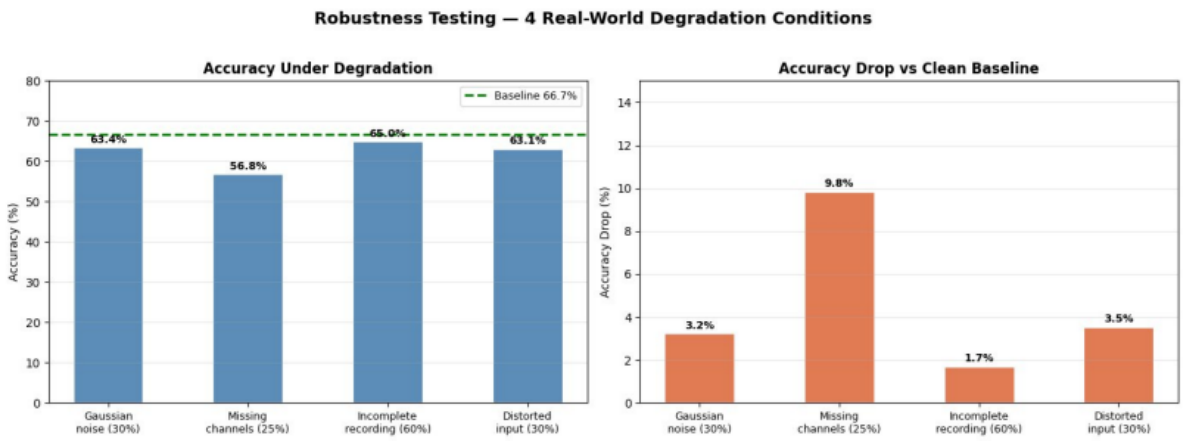

Figure 3: Robustness testing results

Fig. 3: Robustness testing results. *Left*: ensemble accuracy under each degradation condition relative to the clean baseline (66.7%). *Right*: absolute accuracy drop per condition—electrode failure causes the largest degradation (9.8%), while incomplete recording has minimal impact (1.7%). Figure 3 shows how model performance degrades under different real-world signal conditions, highlighting the critical impact of missing EEG channels.

e Cognitive State Distribution

The test-set cognitive state distribution (Figure a, bottom right) is: Positive 2,226 windows (29.0%), Fatigued 2,086 (27.2%), Stressed 1,745 (22.7%), and Calm 1,623 (21.1%). Each of the four states is meaningfully represented: the model does not degenerate to a degenerate prediction strategy. The small dominance of Positive and Fatigued states is a natural result of the stimulus design of DEAP: music videos are more likely to induce high-arousal modes of engagement, and the valence scores distribution around the 3.08 mark naturally results in more windows on both sides of the valence boundary in the High Arousal quadrants and Fatigued Low-Arousal quadrant.

f Limitations

There are three weaknesses of the existing system which deserve to be mentioned. To begin with, in this project, valence is not predicted based on EEG, it is obtained using DEAP ground-truth self-report labels. To create a completely autonomous cognitive state detection system would entail an additional, validated valence prediction model trained on EEG characteristics - which is an even more difficult problem yet a current research difficulty. Second, the 32 DEAP subjects were combined to be trained in a subject-independent set- up. This is the harder problem methodologically; models specific to individual subjects would be more accurate, but would have to be calibrated on a case-by-case basis at the time of deployment. Third, the system uses batch mode over pre-segmented windows instead of streaming inference, which constrained its use to real-time BCI applications.

VI. Conclusion

This project explored the question of whether cognitive state detection using EEG would be reliable when faced with four realistic signal degradation conditions. Six weighted classifiers (SVM, Random Forest, LDA, CNN, LSTM and a CNN-LSTM hybrid) that were trained on the DEAP dataset and a sealed test set were tested on clean and degraded conditions. The ensemble obtained 66.7 percent accuracy on clean data and 63-65% accuracy with Gaussian noise, truncation, and corruption of the feature matrices. The worst-case scenario, namely, the failure of the electrode, led to a decrease of 9.8% points to 56.8 percent, a still higher probability than the baseline. The Circumplex Model of Russell was used to make arousal predictions to obtain four cognitive states in a reasonable proportion, and no one state collapsed to one.

The key empirical conclusion is that EEG-based detection of cognitive states is significantly more resistant to loss of temporal signals as compared to loss of spatial electrodes. To practitioners developing real-world BCI implementations, this can be translated into design advice: a high-quality electrode contact and impedance monitoring is worth the money since the cost of losing spatial brain signal is far greater than the cost of losing a few seconds of recording.

The future work is prioritized in three directions. First, real-time streaming inference with a lightweight subset of the ensemble, which probably includes the LDA and a single convolutional model, to support online cognitive state monitoring should be introduced. Second, to obtain a subject-independent generalization estimate with variance bounds, cross-validation of DEAP was done, leaving out a single subject, and then cross-validation was repeated on the remaining 32 subjects. Third, the inclusion of a valence prediction head that is trained on EEG features and does not rely on self-report labels will allow a completely autonomous mapping of cognitive states.

REFERENCES

[1] S. Koelstra *et al.*, "DEAP: A database for emotion analysis using physiological signals," *IEEE Trans. Affective Comput.*, vol. 3, no. 1, pp. 18–31, Jan. 2012. [Online]. Available: https://ieeexplore.ieee.org/document/5871728

[2] S. Koelstra and I. Patras, "DEAP: A Database for Emotion Analysis Using Physiological Signals Dataset Access," Queen Mary University of London, 2011. [Online]. Available: https://www.qmul.ac.uk/eecs/

[3] J. A. Russell, "A circumplex model of affect," *J. Pers. Soc. Psychol.*, vol. 39, no. 6, pp. 1161–1178, 1980. [Online]. Available: https://psycnet.apa.org/doiLanding?doi=10.1037/h0077714

[4] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735-1780, Nov. 1997. [Online]. Available: https://direct.mit.edu/neco/article/9/8/1735/6109/Long-Short-Term-Memory

[5] P. D. Welch, "The use of fast Fourier transform for the estimation of power spectra," *IEEE Trans. Audio Electroacoust.*, vol. 15, no. 2, pp. 70–73, Jun. 1967. [Online]. Available: https://ieeexplore.ieee.org/document/1161901

[6] Y. Li *et al.*, "From regional to global brain: A hierarchical spatial-temporal neural network for EEG emotion recognition," *IEEE Trans. Affective Comput.*, vol. 13, no. 2, pp. 568–578, Apr. 2022. [Online]. Available: https://ieeexplore.ieee.org/document/8736804

[7] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: https://jmlr.org/papers/v12/pedregosa11a.html

[8] P. Virtanen *et al.*, "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nat. Methods*, vol. 17, pp. 261–272, 2020. [Online]. Available: https://doi.org/10.1038/s41592-019-0686-2

[9] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands for EEG-based emotion recognition," *IEEE Trans. Auton. Ment. Dev.*, vol. 7, no. 3, pp. 162–175, Sep. 2015. [Online]. Available: https://ieeexplore.ieee.org/document/7104132

[10] M. Soleymani *et al.*, "Analysis of EEG signals and facial expressions for emotion detection," *IEEE Trans. Affective Comput.*, vol. 7, no. 1, pp. 17–28, Jan. 2016. [Online]. Available: https://ieeexplore.ieee.org/document/7112127

[11] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273-297, Sep. 1995. [Online]. Available: https://link.springer.com/article/10.1007/BF00994018

[12] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5-32, Oct. 2001. [Online]. Available: https://link.springer.com/article/10.1023/A:1010933404324

[13] V. J. Lawhern *et al.*, "EEGNet: A compact CNN for EEG-based BCIs," *J. Neural Eng.*, vol. 15, no. 5, p. 056013, 2018. [Online]. Available: https://iopscience.iop.org/article/10.1088/1741-2552/aace8c

[14] X. Li *et al.*, "Exploring EEG features in cross-subject emotion recognition," *Front. Neurosci.*, vol. 12, p. 162, 2018. [Online]. Available: https://www.frontiersin.org/articles/10.3389/fnins.2018.00162/full

[15] S.-E. Moon, S. Jang, and J.-S. Lee, "CNN approach for EEG-based emotion recognition," in *Proc. IEEE ICASSP*, pp. 2556–2560, 2018. [Online]. Available: https://ieeexplore.ieee.org/document/8461315